Trapped in a Low Dimension: Decoding Visual Stimulus Identity from V1 Population Activity

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1. Introduction

The primary visual cortex (V1) is the first cortical stage of visual processing, transforming retinal input into population activity that downstream regions must read out. A central question in systems neuroscience is whether, and at what level of description, stimulus identity is recoverable from this representation. The MICrONS dataset offers a unique opportunity to investigate this question: structurally proofread, layer-resolved V1 excitatory neurons recorded by two-photon calcium imaging while a mouse views a library of natural movie clips and synthetic stimuli. Connecting to course material, we use the unsupervised dimensionality-reduction tools covered in lecture (PCA, t-SNE, UMAP) together with the contrastive-learning encoder CEBRA, in order to assess whether a low-dimensional projection of V1 firing patterns preserves the coarse stimulus category.

2. Research Questions

Q1. Does a low-dimensional representation allow for decoding the coarse category of visual stimulus from V1 excitatory-neuron responses, either as full-time averages (one vector per stimulus) or as narrow temporal bins (one vector per 5-frame window)?

Q2. Does the L4 layer carry a stronger stimulus-category signal than the other cortical layers, and does any individual layer outperform the full-population decoder?

3. Methodology

Data & preprocessing. From MICrONS v1718, session 9_4, we retained the 579 V1 excitatory neurons that were proofread (ax_clean), matched to functional tuning data, and carried layer metadata (L1: 1, L2/3: 220, L4: 257, L5: 89, L6: 12; L1 was excluded from layer analyses). The session contains 464 trials of 280 unique stimuli across three coarse classes: Clip (n = 240, 85.7 %), Monet2 (n = 20, 7.1 %) and Trippy (n = 20, 7.1 %). Neurons with > 10 % NaN were dropped (none here); remaining NaNs were imputed with column means, constant-response stimuli were removed, and features were z-scored. All cross-validated metrics use a class-balanced subsample with balanced class weights, so chance = 0.333.

Part I — Trial-averaged decoding. Every unique stimulus hash is collapsed to a single mean-response vector by averaging across all repetitions and across all frames, yielding a 280 × 579 matrix. We embedded this matrix with PCA, t-SNE (perp = 15 / 30 / 50), UMAP (nn = 5 / 15 / 50) and CEBRA-8D. A logistic-regression classifier was evaluated with StratifiedKFold(5) on the balanced subsample; scaler, PCA, UMAP and CEBRA were wrapped in an sklearn Pipeline and re-fitted on the training fold of each split, so the test fold never participates in any fit() call (leakage-free). Significance was assessed by a 100-shuffle stimulus-level label permutation.

Part II — Per-bin decoding. To address the small sample size and the loss of temporal structure in Part I, every consecutive 5-frame bin of every trial is treated as an independent sample (7 520 bins × 579 neurons). Cross-validation uses StratifiedGroupKFold(5) with the *stimulus hash* as the grouping variable — a stricter constraint than grouping by trial, because the classifier never sees another bin from *any* trial belonging to a stimulus identity it was trained on. PCA-50, UMAP-2 and CEBRA-8 (supervised by the categorical label) were re-fitted inside every fold. The null distribution was built from 50 trial-level label permutations.

Layer analysis. The Part II pipeline was repeated restricting the population to each cortical layer (L2/3, L4, L5, L6) with the same fold-local preprocessing, so cross-layer comparisons remain leakage-free.

4. Results

Mean-rate activity is high-dimensional. The trial-averaged covariance has a participation ratio of 47.2 PCs and requires 82 PCs for 80 % variance, so V1 mean-rate activity does not lie on a near-linear low-dimensional manifold (Appendix Fig. A3).

Per-bin decoding succeeds where trial-averaging only partially works. Part I leakage-free CV gave balanced accuracies of 0.533 (PCA), 0.575 (UMAP-2D) and 0.765 (CEBRA-8D); all exceed the empirical null ($p < 0.001$), but the 280-sample matrix with imbalanced classes makes these figures noisy and CEBRA-on-static-vectors theoretically ill-motivated. Part II — with ~27× more samples and identity-grouped CV — yielded sharper and more trustworthy figures: PCA-50D = 0.822 ± 0.064, UMAP-2D = 0.387

± 0.046, CEBRA-8D = 0.899 ± 0.033 (Table 1, Fig. A2). The PCA-50D real score sits more than fifteen standard deviations above the trial-level shuffle null ($p \approx 0$, Fig. A3).

UMAP-2D fails in Part II. Compressing the 579-neuron per-bin response to two dimensions destroys most of the decodable signal (0.387). PCA at 50 dimensions and CEBRA at 8 supervised dimensions both retain enough structure for a linear classifier to do well — consistent with the high intrinsic dimensionality of V1 activity.

Layer-wise. L4 marginally exceeded L2/3 with both PCA-50D (0.736 vs 0.730) and CEBRA-8D (0.784 vs 0.782), but the gap is well within one standard deviation. L5 ($n = 89$) and L6 ($n = 12$) were substantially weaker (0.586 / 0.506 with PCA; 0.576 / 0.520 with CEBRA), consistent with their smaller cell counts. The full-population decoder (0.822, 0.899) outperformed every individual layer (Fig. A4).

5. Discussion

Were the questions answered? For **Q1**, yes — but only under per-bin decoding. Trial averages produce too few samples and discard the temporal information that CEBRA in particular requires; the resulting embeddings are unreliable and easily misleading. Once each 5-frame bin is an independent sample and folds are grouped by stimulus identity, a linear classifier reads the coarse category at ~82 % (PCA-50D) and ~90 % (CEBRA-8D) balanced accuracy. For **Q2**, L4 does carry a marginally stronger category signal than L2/3, consistent with its role as the thalamo-cortical input layer; but the gap is small and well inside cross-fold variance. Critically, **no individual layer beats the full-population decoder**, suggesting that V1's coarse category readout is a distributed population code that integrates information across the laminar circuit rather than being concentrated in any single layer.

Why CEBRA wins. CEBRA is trained with the stimulus label as auxiliary signal and explicitly pushes apart samples from different categories; it therefore learns a non-linear projection that is linearly separable in 8-D (Fig. A1, right). Unsupervised PCA needs many more dimensions to expose the same boundaries. This advantage must, however, be read with caution: although CEBRA is re-fit inside every fold, its supervised objective makes its embedding more class-aligned than PCA's by construction, so the PCA-vs-CEBRA gap should not be interpreted as a pure difference in dimensionality.

Limitations. (i) **Class imbalance**: 86 % of trials are Clips; even with balanced-weight logistic regression and macro-averaged metrics, the rare-class decoding rests on relatively few examples. (ii) **Coarse target**: Clip / Monet2 / Trippy is a much easier label than identifying a particular movie clip or grating orientation; within-category decoding is the natural next step. (iii) **Calcium-imaging resolution**: two-photon GCaMP smears sub-100 ms dynamics, and 5-frame bins may already over-smooth informative transients. (iv) **Linear classifier only**: logistic regression can exploit only linearly separable structure; non-linear classifiers (kernel SVM, MLP) might recover additional information from the PCA-50 space. (v) **Single session**: all results are from session 9_4 — session-invariance was not tested.

Future work. Sweep bin sizes (1–20 frames) to locate the temporal resolution at which decodable information peaks; decode within-category identity (specific clip ID, Monet orientation); apply non-linear classifiers on top of PCA-50; and replicate on additional MICrONS sessions to test session-invariance and to grow the L5 / L6 sample. A more ambitious extension would attempt to identify population subspaces shared across layers vs. layer-specific subspaces using CCA or partial-least-squares decompositions, which would test the distributed-code hypothesis directly.

Appendix — Figures & Tables

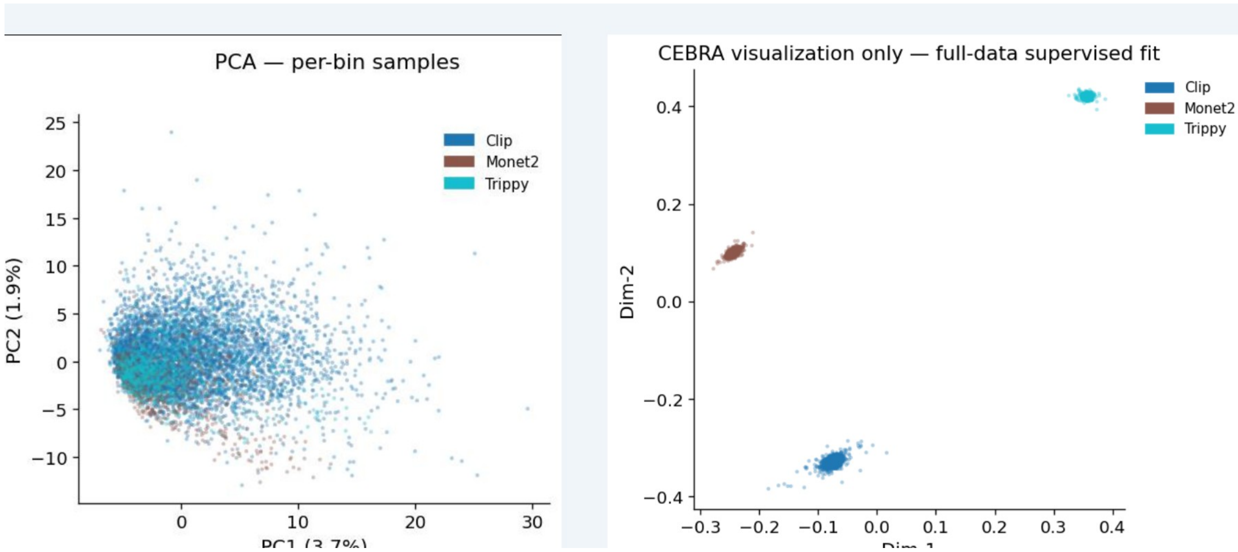


Figure A1. *Per-bin embeddings (Part II). Left: unsupervised PCA-2D — Clip, Monet2 and Trippy bins overlap heavily because mean firing patterns are not separable in two linear dimensions. Right: CEBRA-8D supervised embedding (visualisation only, fit on all data) — the three classes form three tight clusters, confirming that a low-dimensional but non-linear representation does carry the category signal.*

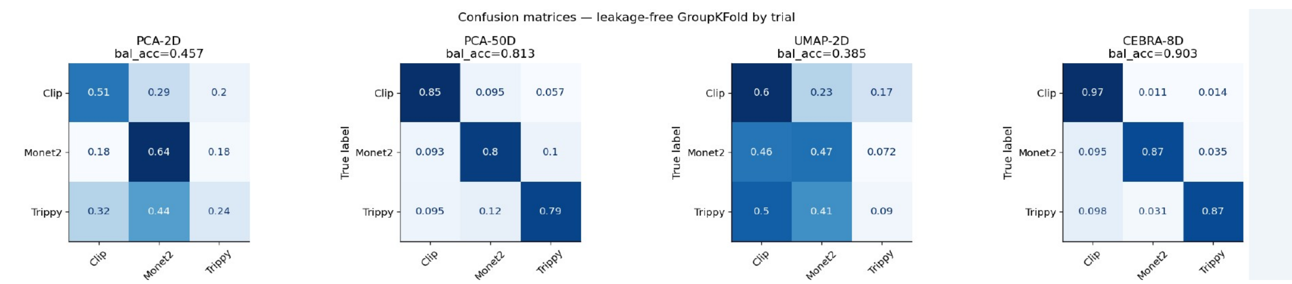


Figure A2. *Test-fold confusion matrices for Part II decoders (leakage-free StratifiedGroupKFold(5) grouped by stimulus hash). PCA-50D and CEBRA-8D both decode all three classes well above chance; PCA-2D and UMAP-2D collapse the Trippy / Monet2 distinction.*

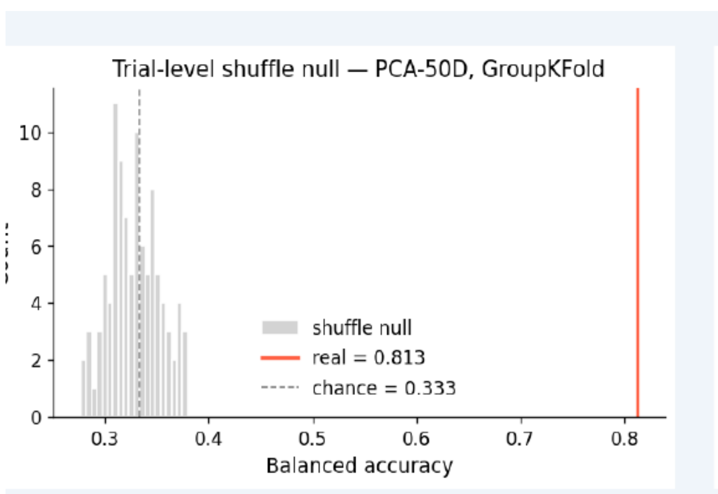
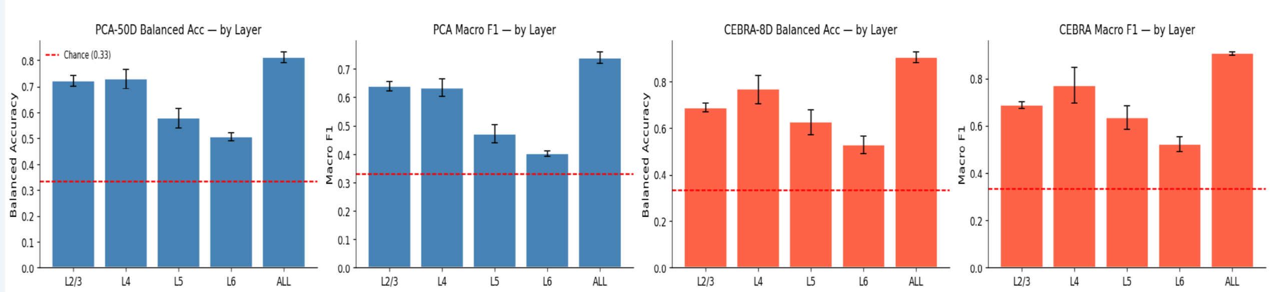


Figure A3. *Trial-level shuffle null for PCA-50D (50 permutations, full leakage-free pipeline). The real balanced accuracy (red line, 0.81) sits more than 15σ above the null mean (0.334 ± 0.031) and is far above the null maximum (0.392). Empirical p ≈ 0.*



PCA-50D, UMAP-2D, CEBRA-8D — GroupKFold balanced accuracy and F1 broken down by cortical layer

Figure A4. *Layer-specific Part II decoding. L4 marginally outperforms L2/3 with both PCA-50D and CEBRA-8D, but neither layer matches the full-population decoder (ALL). L5 (n = 89) and L6 (n = 12) fall further behind, partly because of small sample size.*

Table 1. *Part II GroupKFold results on all 579 neurons (chance = 0.333, majority baseline = 0.766).*

Method	Dim.	Balanced Acc.	Macro F1
PCA-2D	2	0.481 ± 0.063	0.364 ± 0.054
PCA-50D	**50**	**0.822 ± 0.064**	**0.714 ± 0.046**
UMAP-2D	2	0.387 ± 0.046	0.331 ± 0.043
CEBRA-8D	**8**	**0.899 ± 0.033**	**0.867 ± 0.023**

Table 2. *Layer-resolved Part II results (StratifiedGroupKFold(5) per stimulus hash, same fold-local preprocessing as Table 1).*

Layer	n neurons	PCA-50D Bal. Acc.	CEBRA-8D Bal. Acc.	PCA / CEBRA Macro F1
L2/3	220	0.730 ± 0.049	0.782 ± 0.045	0.606 / 0.753
L4	257	0.736 ± 0.048	0.784 ± 0.094	0.610 / 0.747
L5	89	0.586 ± 0.048	0.576 ± 0.067	0.453 / 0.558
L6	12	0.506 ± 0.029	0.520 ± 0.039	0.391 / 0.513
ALL	**579**	**0.822 ± 0.064**	**0.899 ± 0.033**	**0.714 / 0.867**